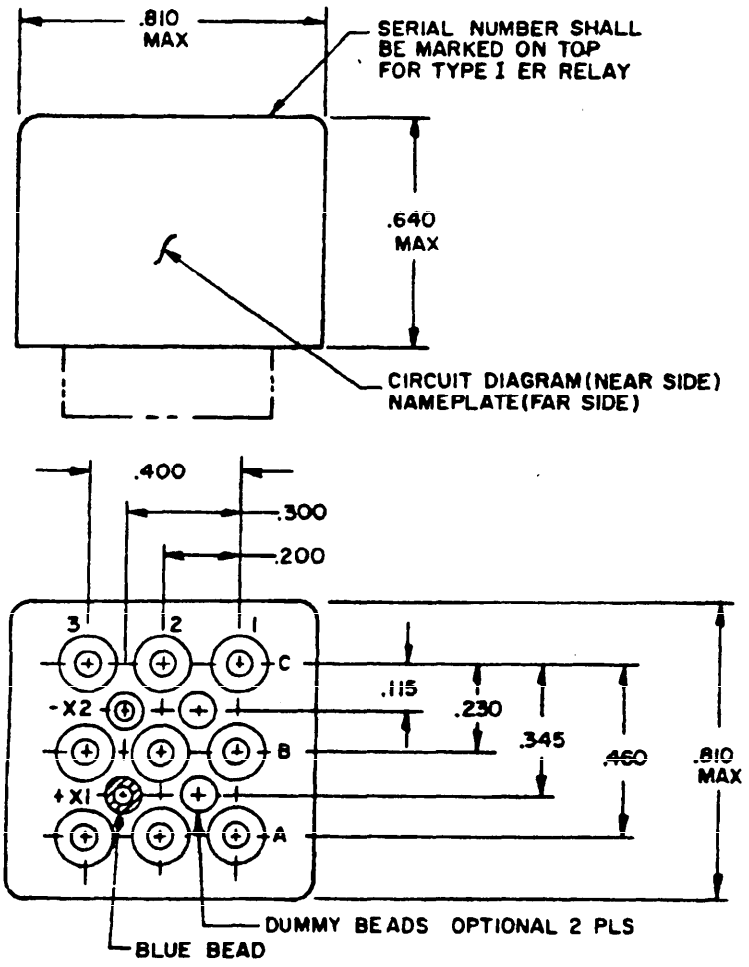
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES), PERMANENT MAGNET DRIVE, LOW LEVEL TO 10 AMPERES, 3 PDT, ALL WELDED, HERMETICALLY SEALED

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.

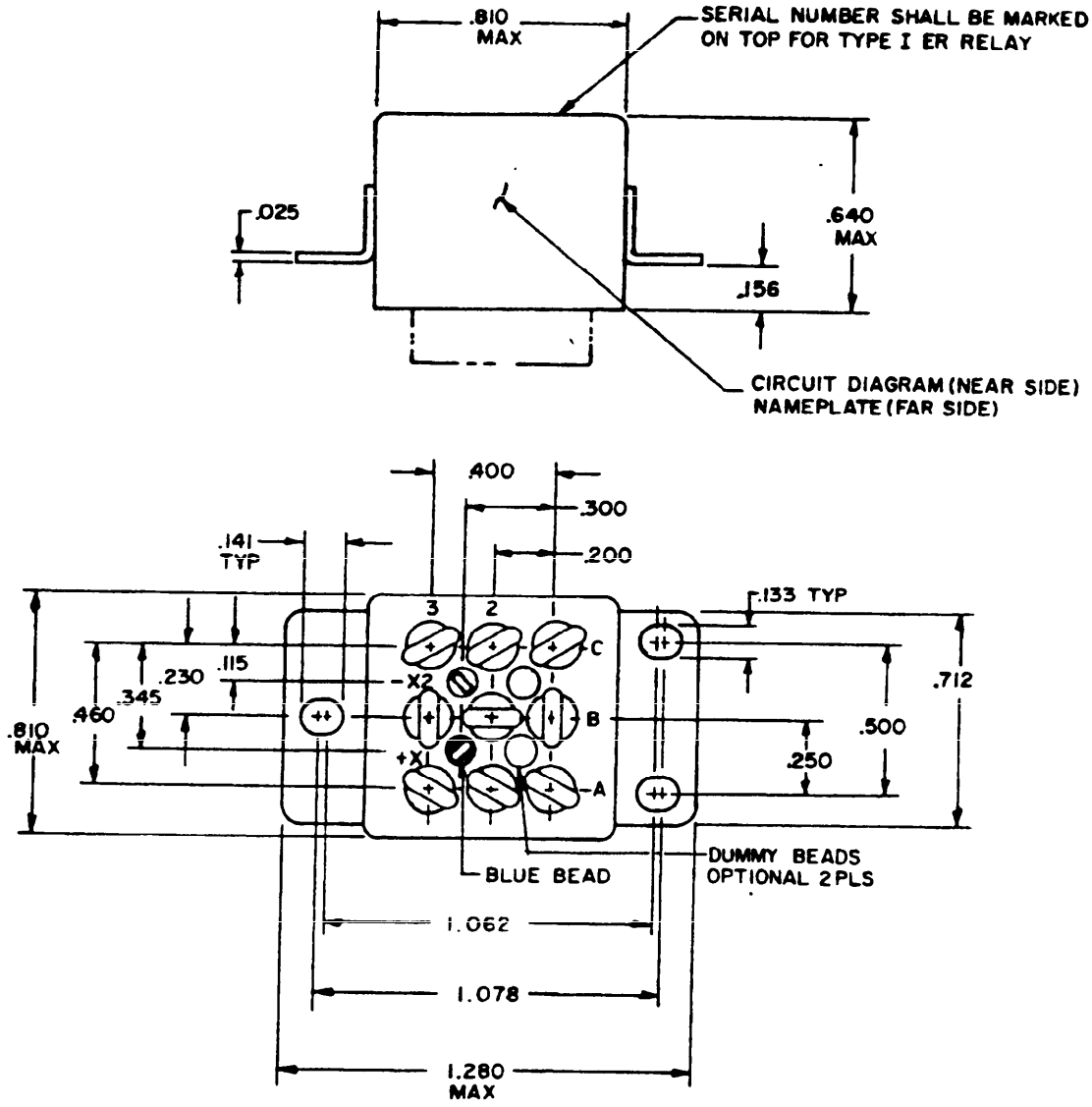


CONFIGURATION A

FIGURE 1. Outline drawing (for details, see table I).

(B) denotes changes

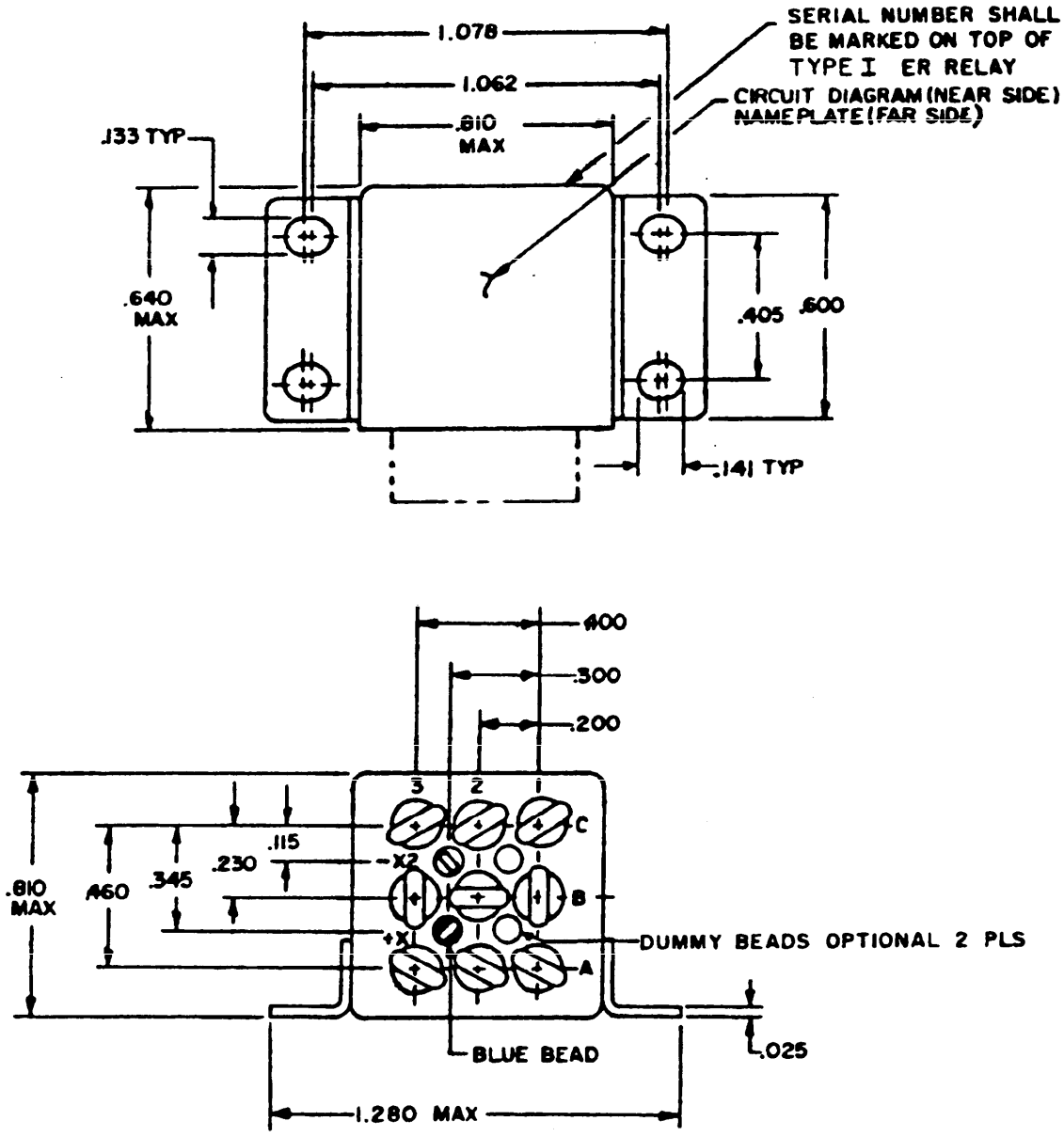
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CONFIGURATION B

FIGURE 1. Outline drawing (for details, see table I) - Continued.

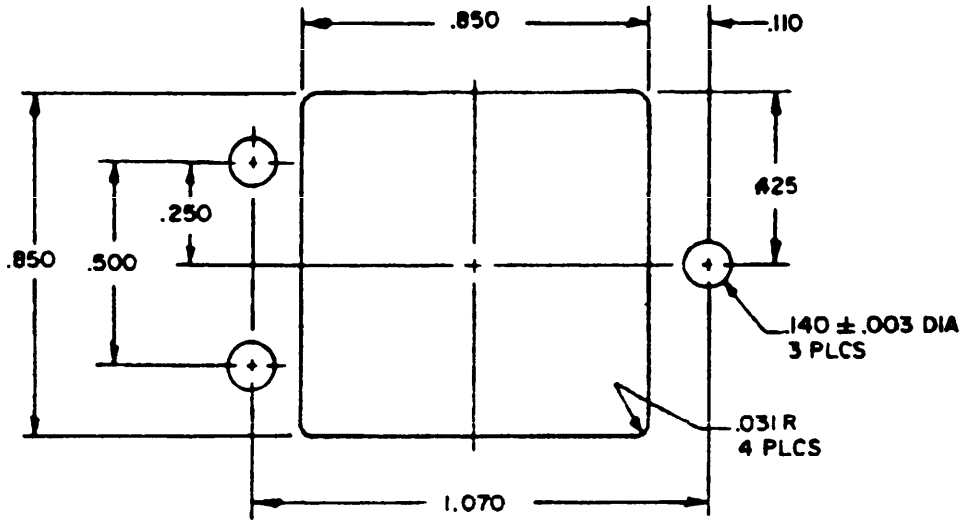
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CONFIGURATION C

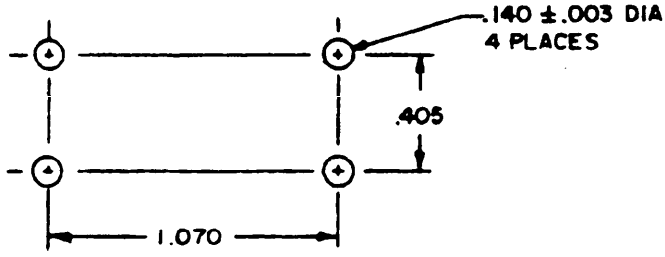
FIGURE 1. Outline drawing (for details, see table I) - Continued.

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MOUNTING HOLE LAYOUT

CONFIGURATION B

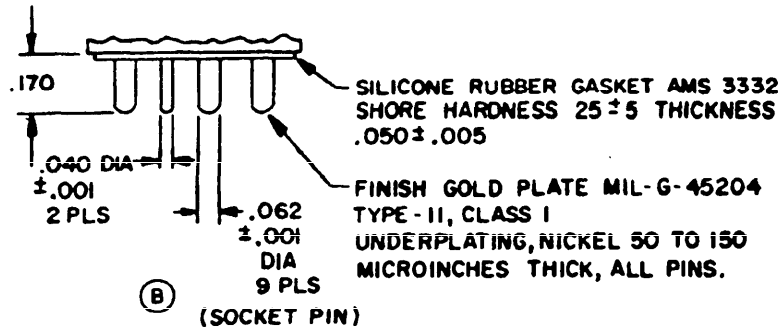


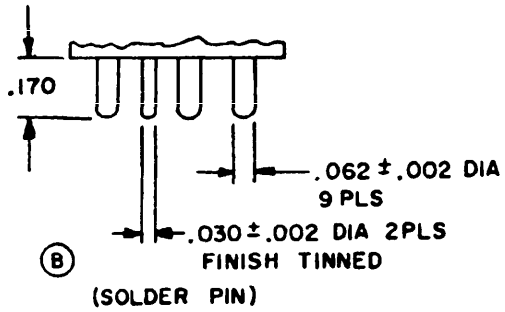
MOUNTING HOLE LAYOUT

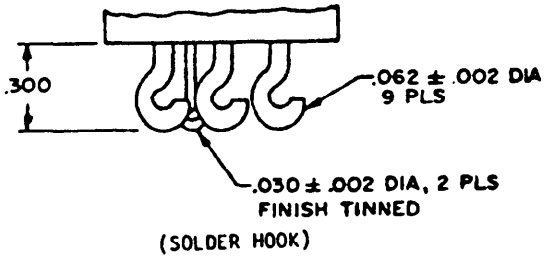
CONFIGURATION C

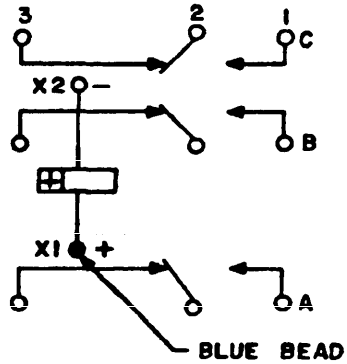
FIGURE 1. Outline drawing (for details, see table I) - Continued.

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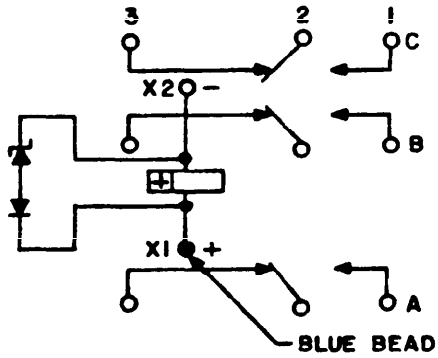








TYPE A



TYPE B

CIRCUIT DIAGRAM

(B)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	0.03	.110	2.79	.230	5.84	.600	15.24
.002	0.05	.115	2.92	.250	6.35	.640	16.26
.003	0.08	.133	3.38	.300	7.62	.712	18.08
.025	0.64	.140	3.56	.345	8.76	.810	20.57
.030	0.76	.141	3.58	.400	10.16	.850	21.59
.031	0.79	.156	3.96	.405	10.29	1.062	26.97
.040	1.02	.170	4.32	.425	10.80	1.078	27.38
.050	1.27	.200	5.08	.500	12.70	1.280	32.51

NOTES:
1. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
2. Metric equivalents are given for general information only.
3. Dimensions are in inches.
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux.

FIGURE 1. Outline drawing (for details, see table I) - Continued.

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Ⓑ TABLE I. Mechanical and physical characteristics. 1/

Military part number 2/ 3/	Type	Coil suppression (transient voltage, back EMF) 2/	Terminal type	Mating socket	Mounting	Circuit type
M6106/29-001	I	N/A	Solder pin	N/A	Configuration A	A
" -002	I ER	N/A	"	"	"	A
" -003	I	42 V dc max	"	"	"	B
" -004	I ER	42 V dc max	"	"	"	B
" -005	I	N/A	Solder hook	"	"	A
" -006	I ER	N/A	"	"	"	A
" -007	I	42 V dc max	"	"	"	B
" -008	I ER	42 V dc max	"	"	"	B
" -009	I	N/A	Socket pin		"	A
" -010	I ER	N/A	"	MIL-S-12883/	"	A
" -011	I	42 V dc max	"	46-01	"	B
" -012	I ER	42 V dc max	"		"	B
" -013	I	N/A	Solder pin	N/A	Configuration B	A
" -014	I ER	N/A	"	"	"	A
" -015	I	42 V dc max	"	"	"	B
" -016	I ER	42 V dc max	"	"	"	B
" -017	I	N/A	Solder hook	"	"	A
" -018	I ER	N/A	"	"	"	A
" -019	I	42 V dc max	"	"	"	B
" -020	I ER	42 V dc max	"	"	"	B
" -021	I	N/A	Socket pin		"	A
" -022	I ER	N/A	"	MIL-S-12883/	"	A
" -023	I	42 V dc max	"	46-01	"	B
" -024	I ER	42 V dc max	"		"	B
" -025	I	N/A	Solder pin	N/A	Configuration C	A
" -026	I ER	N/A	"	"	"	A
" -027	I	42 V dc max	"	"	"	B
" -028	I ER	42 V dc max	"	"	"	B
" -029	I	N/A	Solder hook	"	"	A
" -030	I ER	N/A	"	"	"	A
" -031	I	42 V dc max	"	"	"	B
" -032	I ER	42 V dc max	"	"	"	B

1/ For type I ER relays, a suffix letter, M, U, X, or P to indicate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations), M = 1%, U = 0.5%, X = 0.3% and P = 0.1%. Example: 001M ----, 003P.

2/ JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression.
Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.
Diodes shall have a peak inverse voltage of 600 V dc minimum.

3/ M6106/29-033 through -036 are canceled without replacement.

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REQUIREMENTS:

Coil data:

Coil data: See tables I and II.

Duty rating: Continuous.

Operational data:

Rated contact load (endurance): See table III.

Operate time: See table II.

Release time: See table II.

Break bounce: See table II.

Circuit breaker compatibility: Applicable. 1/

Physical:

Dimensions and configuration: See figure 1.

Weight (maximum): .062 pound, 28.15 grams.

Terminal type: See figure 1 and table I.

ENVIRONMENTAL CHARACTERISTICS:	Configurations A and B	Configuration C
Temperature rating	-70°C to +125°C	-70°C to +125°C
Maximum altitude rating	300,000 feet	300,000 feet
Shock g-level	200 g	100 g
Duration	6 ms	6 ms
Maximum duration contact opening	10 µs	10 µs
Vibration - sinusoidal:		
G-level	30 g	20 g
Frequency range	70 - 3,000 Hz	57 - 3,000 Hz
Vibration - random:		
Applicable standard	MIL-STD-202	MIL-STD-202
Method 214	0.4 g^2/Hz	0.2 g^2/Hz
Test condition	IG	IE
Duration	15 min ea plane	15 min ea plane
Acceleration	15 g	15 g

ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1,250 V rms	1,000 V rms
Aux contacts	---	---
(B) All other points	1,250 V rms	1,000 V rms

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence as listed in table IV. Separate relays shall be tested at 28 V dc and 115/200 V ac 400 Hz 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

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Ⓑ TABLE II. Operating characteristics.

Dash number	Coil	Coil data								Drop 1/ out voltage	Hold voltage 1/	Time milliseconds (maximum) 2/						
		Nominal			Max		Max pick-up voltage 1/							Contact bounce				Break bounce NO only
		Volts	Freq. Hz	Res. Ω±10% @25°C	Volts	Amp	Normal	High temp test	Cont current test			Operate	Release	Main NO	Main NC	Aux NO	Aux NC	
-001 thru -032	X1 X2	28	DC	400	29	.081	18	19.8	22.5	1.5	7.0	6	6	1	1	--	--	.1

1/ Over temperature range.
2/ With nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	10	10	--	--	10	--	--	--	10	--	--	--	
Inductive	10	6	6	--	--	--	--	--	--	--	--	--	--	
Inductive	20	--	--	--	--	8	--	--	--	8	--	--	--	
Motor	50	4	4	--	--	4	--	--	--	4	--	--	--	
Lamp	50	2	2	--	--	2	--	--	--	2	--	--	--	
Transfer load	--	--	--	--	--	--	--	--	--	--	--	--	--	1/
Mechanical life reduced current	200	1.25	.125	--	--	1.25	--	--	--	1.25	--	--	--	
Min current	Applicable per specification													
Low level	100	Applicable per specification												2/ 3/ 4/
Multiple loads	As applicable													5/

1/ Transfer load indicates relay suitable for transfer between unsychronized ac power supplies at rating indicated.
2/ For low level the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10 to 50 millivolt open circuit voltage, 100 ohm maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.
3/ For ER relays, only the alternate low level test of operational reliability shall be used.
4/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.
5/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

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Dielectric withstanding voltage (altitude): 2/

	80,000 feet	300,000 feet
Coil to case	250 V rms	500 V rms
Aux contacts	---	---
All other points	250 V rms	500 V rms

Maximum contact drop (initial): 0.150 volt.

After life test: 0.175 volt.

Overload current: 30 amperes dc, 60 amperes ac.

Rupture current: 40 amperes dc, 80 amperes ac.

Part number: M6106/29- (dash number from table I and a suffix letter indicating the failure rate level).

Ⓑ TABLE IV. Time-current relay characteristics at +25°C.

1	20A	1 hour
2	30A	20 seconds
3	75A	2 seconds
4	150A	.53 second

CAUTION: Compare with time-current characteristics of the associated circuit protective device.

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

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Ⓑ TABLE V. Qualification by similarity. 1/

Military part number M6106/29-	Loads		Dynamics			Environmental
	Basic qual	ER	Sockets/Pins			
	A	D	A	B	C	A
	2/ DC 3/	2/ DC 3/	Configuration 2/ A 3/	Configuration 2/ B 3/	Configuration 2/ C 3/	2/ DC 4/
001	4	3	2			4
002	4	3	2			4
003	1,2,3	1,2	1			1,2,3
004	2,3	1,2	1			2,3
005	4	3	2			4
006	4	3	2			4
007	1,2,3	1,2	1			1,2,3
008	2,3	1,2	1			2,3
009	4	3	2			4
010	4	3	2			4
011	1,2,3	1,2	1			1,2,3
012	2,3	1,2	1			2,3
013	4	3		2		4
014	4	3		2		4
015	1,2,3	1,2		1		1,2,3
016	2,3	1,2		1		2,3
017	4	3		2		4
018	4	3		2		4
019	1,2,3	1,2		1		1,2,3
020	2,3	1,2		1		2,3
021	4	3		2		4
022	4	3		2		4
023	1,2,3	1,2		1		1,2,3
024	2,3	1,2		1		2,3
025	4	3			2	4
026	4	3			2	4
027	1,2,3	1,2			1	1,2,3
028	2,3	1,2			1	2,3
029	4	3			2	4
030	4	3			2	4
031	1,2,3	1,2			1	1,2,3
032	2,3	1,2			1	2,3

1/ Reference Amendment 1, MIL-R-6106, for guidelines established for determining how relays are to be grouped and ranked within each subgroup.

2/ DC suppressed coils - discretes qualify only discretes; hybrids qualify only hybrids.

3/ Terminals - socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity.

4/ Socket pins (with gasket seals) are considered nonsimilar to solder pins and hooks and must be tested separately.

Custodians:
Air Force - 85
Navy - EC

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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